Week 4 Lab

Machine Learning Lab

Question 1: Load the Iris dataset using *sklearn.datasets*. Split the data into 70% training and 30% testing. Implement a k-Nearest Neighbors classifier. Experiment with $k = 1, 5$, and 15.

- **Task:** Report the accuracy for each k. Which value performed best and why? At what point does a high k start to "blur" the boundaries too much?

Question 2: Using the Wine dataset from *sklearn.datasets*, train a k-NN model. First, train it on the raw data. Then, apply StandardScaler to the features and retrain.

- **Task:** Compare the accuracy scores. Explain why distance-based algorithms like k-NN are sensitive to the scale of input features.

Question 3: The choice of "distance" defines what the model considers "close." Using the Wine dataset:

- **Task:** Compare metric='*euclidean*' (L2 norm) and metric='*manhattan*' (L1 norm).
- **Calculation:** Explain how the model calculates the distance for a single test point in both cases using the formulas:
    - **Euclidean:** $d(x, y) = \sqrt{\{\sum (x_i - y_i)^2\}}$
    - **Manhattan:** $d(x, y) = \sum |x_i - y_i|$

Question 4: Create a synthetic classification dataset using *make_classification*. Train two Decision Tree models: one using criterion='*gini*' and the other using criterion='*entropy*'.

- **Task:** Visualize both trees using *plot_tree*. Identify one node where the split differs and explain the mathematical difference between Gini Impurity and Information Gain (Entropy).
- **Task:** Visualize the first two levels of each tree. Do they pick the same feature for the root split?

Question 5: Decision trees can handle both "Yes/No" data and numbers.

- **Task:** Create a small dummy DataFrame (e.g., Features: 'Is it Raining?', 'Temperature'; Target: 'Go to Park?').
- **Task:** Use $LabelEncoder$ to turn the "Yes/No" into 0s and 1s, then train a tree. Explain how the tree treats a "1" vs a temperature of "90°F."

Question 6: Decision Trees are notorious for overfitting. Using the Breast Cancer dataset, train a tree with no depth limit. Then, train another tree with max_depth=3 and min_samples_leaf=5.

- **Task:** Compare the training accuracy vs. testing accuracy for both. Explain the "bias-variance tradeoff" in the context of tree depth.

Question 7: The Perceptron is the simplest neural network. Use NumPy to generate two sets of points that are linearly separable and two sets that are not (e.g., an XOR pattern).

- **Task:** Attempt to train *sklearn.linear_model.Perceptron* on both. Use a scatter plot to show the decision boundary. Why does the Perceptron fail on the second dataset?

Question 8: Forget Scikit-Learn for a moment. Implement the Perceptron learning algorithm from scratch using only NumPy.

- **Task:** Your function should update weights using the rule:

$$w \leftarrow w + \eta(y^i - \hat{y}^i)x^i$$

Train it on a toy binary dataset and plot the weight convergence over 50 epochs.

Question 9: Using a simple 2D dataset, train a Linear SVM (*SVC(kernel='linear')*).

- **Task:** Plot the decision boundary and highlight the support vectors. How many support vectors were chosen, and how do they define the margin?

Question 10: Using a dataset with significant overlap (noise), train multiple SVMs with C values of 0.01, 1, and 100.

- **Task:** Observe the margin width and the number of misclassified points in the training set for each. Describe the relationship between the regularization parameter C and the model's tolerance for errors.

Question 11: Pick a real-world dataset (e.g., Pima Indians Diabetes or Digits).

- **Task:** Implement k-NN, Decision Trees, Perceptron, and SVM. Perform a 5-fold cross-validation for each and create a table comparing:
    1. Average Accuracy
    2. Training Time
    3. Prediction Speed (Latency)
- **Discussion:** Which model would you choose if you needed a model that makes predictions in microseconds? Which would you choose if interpretability (explaining *why* a choice was made) was the top priority?